OPTIMIZATION USING CALCULUS AND PYTHON

Maximum Area of a Rectangular Field

A beginner-friendly mathematical modelling and Python project

Project Report

Abstract

This project finds the dimensions of a rectangle with the greatest area when the total fencing is fixed at 40 metres. The problem is modelled algebraically, solved using derivatives, and checked with a short Python program and graph.

1. Introduction

Optimization means finding the maximum or minimum of a quantity under given conditions. It is used in engineering, economics, manufacturing, agriculture, and many other areas. This project demonstrates the connection between calculus and programming.

2. Problem Statement

A farmer has 40 metres of fencing and wants to enclose a rectangle. Find the length and width that produce the maximum possible area. Assume all fencing is used for the four sides and both dimensions are positive.

3. Mathematical Model

Let x be the length and y be the width, both in metres. The perimeter constraint is 2x + 2y = 40. Dividing by 2 gives x + y = 20, so y = 20 – x. The area is A = xy. Substituting the constraint gives the objective function:

$A(x) = x(20 - x) = 20x - x^2$

4. Calculus Solution

Differentiate the area function: $A'(x) = 20 - 2x$. Set the derivative equal to zero: $20 - 2x = 0$, so $x = 10$ metres. The second derivative is $A''(x) = -2$, which is negative, confirming a maximum. The width is $y = 20 - 10 = 10$ metres. Therefore, the maximum area is $10 \times 10 = 100$ m².

5. Results

Quantity	Result
Total fencing	40 m
Optimal length	10 m
Optimal width	10 m
Maximum area	100 m²
Shape	Square

6. Python Code

NumPy samples possible lengths and calculates each area. np.argmax identifies the index of the largest sampled area. Matplotlib draws the curve and marks the maximum point.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 20, 101)
area = x * (20 - x)
i = np.argmax(area)

print("Best length:", x[i], "m")
print("Maximum area:", area[i], "m^2")

plt.plot(x, area)
plt.scatter(x[i], area[i], color="red")
```

```
plt.xlabel("Length (m)")
plt.ylabel("Area (m^2)")
plt.title("Maximum Area Optimization")
plt.grid()
plt.show()
```

7. Graph and Interpretation

The graph is a downward-opening parabola. The area rises as length approaches 10 m, reaches 100 m² at x = 10, then falls. Since 101 sample points are used, x = 10 is included exactly.

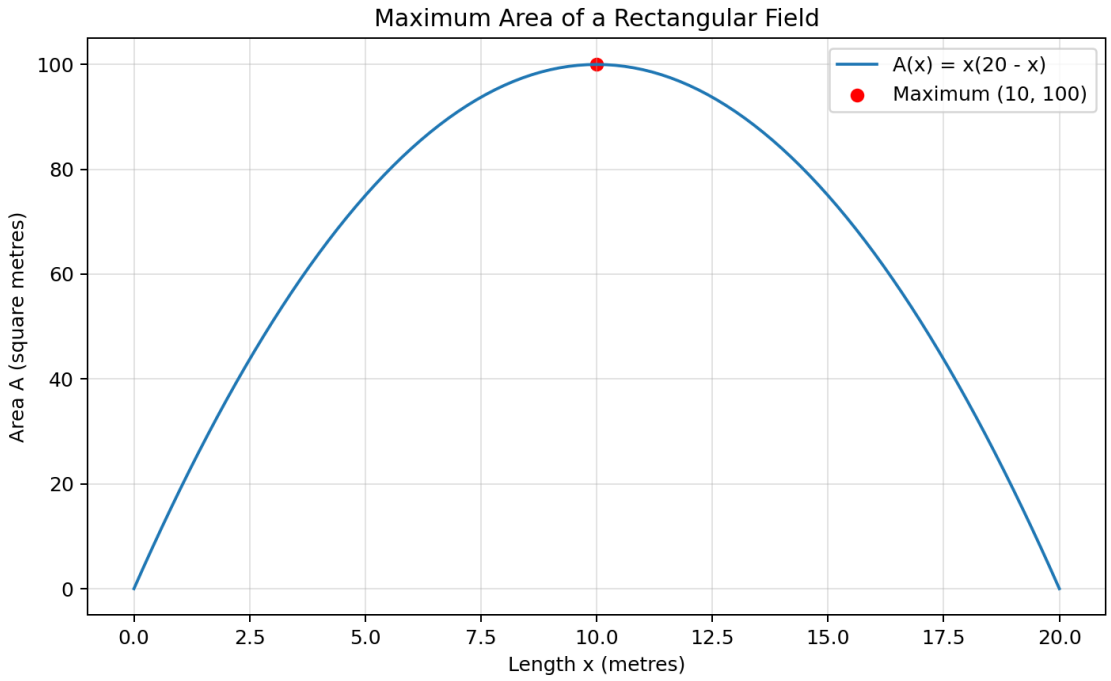


Figure 1. Area as a function of the rectangle's length.

8. Conclusion

For a rectangle with a fixed perimeter of 40 metres, the greatest area in this model is 100 m², obtained with a 10 m by 10 m square. Calculus proves the exact optimum, while Python provides a numerical check and visual explanation.

9. Assumptions and Limitations

The model assumes a flat rectangular field, fixed fencing, and no gate or material loss. It does not include terrain or construction costs. The Python program checks sampled values; the derivative calculation establishes the exact result.

10. How to Run

Install Python 3, then run `python -m pip install -r requirements.txt` in this folder. Run the project with `python optimization.py`.

References

1. Stewart, J., *Calculus: Early Transcendentals*, Cengage Learning (standard calculus methods).

2. NumPy documentation: https://numpy.org/doc/

3. Matplotlib documentation: https://matplotlib.org/stable/